Flood Vulnerability Analysis

Kurla, Mumbai, India

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Executive Summary

The current run processed 2285 buildings and generated dynamic vulnerability classifications.

Study Area

The study area is defined by the OSM building extent and the associated DEM coverage. Distances use projected metric CRSs; map outputs use EPSG:4326.

Data Sources

Source	Use
OpenStreetMap	Buildings and waterways
DEM	D:\MiniProjectTayyabsirDrive\clustering-based-flood-analysis\Project_Files\kurla.tif
K-Means	Unsupervised risk grouping

Feature Summary Statistics

Feature	Mean	Median	Min	Max
elevation	9.440	9.000	-1.500	33.000
dist_to_river	313.962	249.416	0.000	1184.723
slope_deg	86.756	89.989	0.000	89.998
local_relief_m	4.008	4.000	0.000	19.000
dist_to_major_waterway	517.203	376.788	0.000	1735.036
waterway_density_300m	0.001	0.001	0.000	0.007
dist_to_drainage_m	333.373	274.075	0.000	1184.723

Risk Distribution

Risk level	Buildings	Percentage
High Risk	1514	66.258
Medium Risk	82	3.589
Low Risk	689	30.153

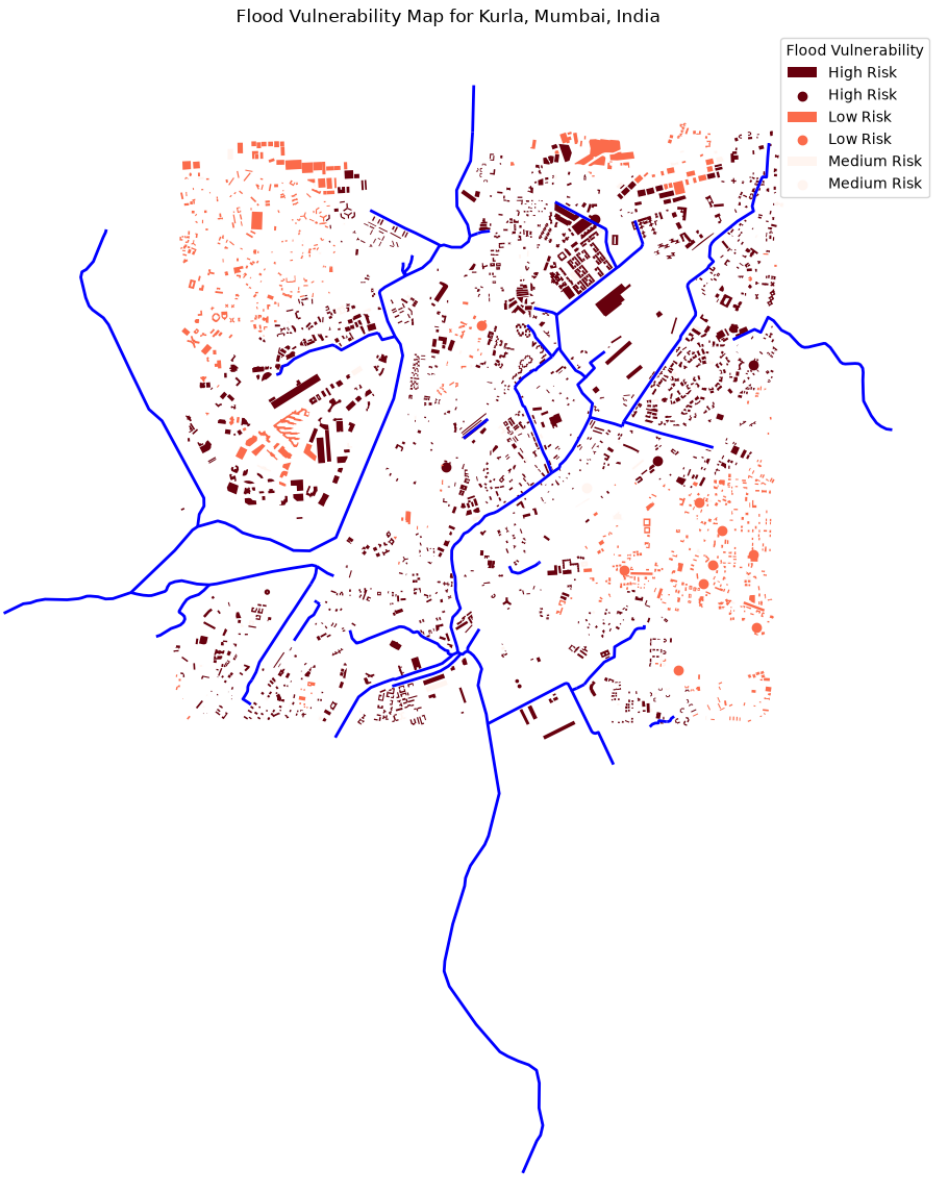
Cluster Profiles

Risk level	elevation	dist_to_river	slope_deg	local_relief_m	dist_to_major_waterway	waterway_density_300m	dist_to_drainage_m
High Risk	7.884	179.216	89.985	3.976	305.124	0.002	205.651
Low Risk	12.937	612.673	89.986	4.312	988.770	0.000	616.448
Medium Risk	8.780	291.919	0.000	2.049	470.614	0.001	313.030

Model Evaluation Metrics

Metric	Value
Silhouette	0.385
Davies-Bouldin	0.925
Calinski-Harabasz	1065.079

Static Risk Map



Insights

- High Risk is the largest risk group, containing 1514 buildings (66.3% of the study buildings).

- The lowest mean elevation occurs in the High Risk group (7.884 m).

- The High Risk group has the smallest average distance to waterways (179.216 m).

Limitations and Future Work

K-Means is unsupervised and labels are assigned by mean elevation. OSM, DEM, and historical-grid coverage and resolution limit interpretation. Validated observations and supervised models are future improvements.

Appendix

Features: elevation, dist_to_river, slope_deg, local_relief_m, dist_to_major_waterway, waterway_density_300m, dist_to_drainage_m

Parameters: K-Means clusters=3, random_state=42, n_init=10; median imputation; standard scaling.